

Woo Kyoung Han

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EDUCATION

(MM/YY~MM/YY)

- Ph.D.** *Korea University, South Korea.* (03/24 ~ Present)
- *Dept. of Electrical and Computer Engineering (Signal Processing and Multimedia)*
- 4.25/4.50 GPA
- Advisor: Kyong Hwan Jin
- M.S.** *Daegu Gyeongbuk Institute of Science & Technology (DGIST), South Korea.* (03/22 ~ 02/24)
- *Dept. of Electrical Engineering and Computer Science*
- 4.00/4.30 GPA
- Advisor: Kyong Hwan Jin & Sunghoon Im
- B.S.** *Soongsil University, South Korea.* (03/16 ~ 02/22)
- *School of Electronic Information Engineering*
- 4.13/4.5 GPA

PUBLICATION

W. K. Han, Y. Kim, J. Kim, and K.H. Jin “Codec-derived Information Priors for Video Token Sparsification in Vision Language Model Inference” *Manuscript Under Review*, 2026.

H. Cho, **W. K. Han**, and K. H. Jin “Balancing Fidelity and Diversity in Diffusion Models via Symmetric Attention Decomposition: Hopfield Perspective” *43rd International Conference on Machine Learning (ICML)*, 2026.

M. Choi, **W. K. Han**, S. Im and K. H. Jin “Linear Recurrent Unit with Semantic Modulation for Image Super-Resolution.” *IEEE/CVF Conference on Computer Vision and Pattern Recognition Findings (CVPR Findings)*, 2026.

W. K. Han*, Y. Lee*, B. Lee, S. H. Park, S. Im, and K.H. Jin “JPEG Processing Neural Operator for Backward Compatibility” *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025.

W. K. Han, B. Lee, H. Cho, S. Im, and K.H. Jin “Towards Lossless Implicit Neural Representation via Bit Plane Decomposition” *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.

D. G. Choi, J. Lee, J. Koo, **W. K. Han**, D. Park, J. Kung, J.Lee, J.H. Yoon “A 65nm 687.5-TOPS/W Drive Strength-based SRAM Compute-In-Memory Macro with Adaptive Dynamic Range for Edge AI applications” *IEEE Asian Solid-State Circuits Conference (A-SSCC)*, 2024.

M. Kim, Y. Lee, **W. K. Han** and K. H. Jin “Learning Residual Elastic Warps for Image Stitching under Dirichlet Boundary Condition” *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2024.

W. K. Han, S. Im, J. Kim, and K. H. Jin “JDEC: JPEG Decoding via Enhanced Continuous Cosine Coefficients.” *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.

W. K. Han, B. Lee, S. H. Park, and K. H. Jin “ABCD: Arbitrary Bitwise Coefficient for De-Quantization.” *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023.

GRANTS

- AI SeoulTech Graduate Scholarship [10,000,000 KRW] (from the Seoul Scholarship Foundation) (2025)
Ph.D. Candidate Fellowship [50,000,000 KRW] (from the National Research Foundation of Korea) (2024~2026)
Merit-based Scholarship (Korea University) (2024)

WORK & RESEARCH EXPERIENCE

- Teaching Assistant* (2018 ~ 2024)
- Electromagnetism, Circuit Theory, Linear Algebra, Fundamental Signal Processing, Advanced Programming (Soongsil University)
 - Engineering Mathematics, Signal Processing. (DGIST)
- Research Assistant* (2022 ~ Present)
- Platform development for the police health Management (Supported by National Police Agency of Korea)
 - Implemented JPEG-domain processing pipelines in PyTorch/C++
 - Improved robustness under distortion scenarios (quantization, noise)
 - Developed medical image compression and restoration modules for a police health monitoring platform

AWARDS

- ICML Reviewer Award – Silver (2026)
- Outstanding Graduate Student Award, Korea University (2026)
- Workshop On Image Processing and Image Understanding – Bronze Prize (2025)
- Workshop On Image Processing and Image Understanding – Encouragement Prize (2024)
- IEIE Summer Annual Conference – Best Student Paper (2024)
- IEEE Seoul Section Best Students Award – Gold (2023)

PATENTS

- Korea patent publication*
- Lossless implicit neural representation via object signal quantization and bitwise decomposition (2024)
 - System for B-Spline Texture Coefficient Estimation and Method for Generating High-Resolution Images Using the Same (2023)
 - Apparatus and Method of Recovering Image Using Arbitrary Bitwise Coefficient Estimation for De-quantization (2023)